University of Colombo, Sri Lanka

UCSC

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I - UCSC AY22 [held in September/October 2025]

SCS1302 - Discrete Mathematics

(Two (2) Hours)

Answer ALL questions

Number of Pages = 3 Number of Questions = 4

To be completed by the candidate	
Index Number:	

244

Important Instructions to candidates:

I. Students should answer in the medium of English language only using the space provided in this question paper.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number CLEARLY on each and every page of this Question paper.

IV. This paper consists of 4 questions in 3 pages (including the Cover Page).

V. Answer ALL questions.

VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.

VII. Non-Programmable calculators are not allowed

VIII. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.

IX. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

1)

a) Which of the following sentences are propositions? What are the truth values of those that are propositions?

i. India is the capital city of Sri Lanka.
ii. $2 + 3 = 5$.
iii. $5 + 7 = 10$.
iv. Good Morning

(08 Marks)

b) Use quantifiers and predicates to express the following statements.

i. Every computer science student needs a course in discrete mathematics.
ii. There is a student in this class who owns a personal computer.

(08 Marks)

c) Give the negation of the statement $\forall x \; \exists y \; (P(x, y) \rightarrow Q(x, y))$

(04 Marks)

d) Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Nimal is a student in this class" imply the conclusion "Nimal has taken a course in computer science."

(05 Marks)

2)

a) Let $A = \{1,3,5\}$ and $B =\{5,8,10\}$. Compute the following.

i) $A \cap B$
ii) $A \cup B$
iii) $A - B$
iv) B' if the universal set is $\{x \mid x \text{ is an integer} <= 10 \}$
v) Power set of A

(10 Marks)

b) Prove that for every set S, $\{\} \subseteq S$

(05 Marks)

c) Prove that if A, B, and C are sets such that $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

(10 Marks)

3)

a) What is a function? What are the properties of a function.

(5 Marks)

b) Which of the following functions are one to one (1-1). Justify your answer.

i) $f(x) = 2x$
ii) $f(x) = x^2$

(10 Marks)

c) What are the properties of functions that have inverse functions?

(04 Marks)

d) Give an example of a function which can have an inverse. What is the inverse of the function that you have given.

(06 Marks)

4)

a) Define a relation in mathematics. Give an example and explain how it can be represented.

(05 Mraks)

b) Does the relation **less that or equal** ($\leqslant$) defined on the set of real numbers is

I. reflexive,
II. Symmetric
III. Transitive

Justify your answers with examples.

(12 Marks)

c) Give an example of an equivalence relation. Prove that the relation you have given is an equivalence relation.

(08 Marks)